

$Cov(ax, bY) = ab Cov(x, Y)$ $Cov(X, Y) = E(XY) - E(X)E(Y)$
 $Cov(X, X) = Var(X)$ $Cov(X+a, Y+b) = Cov(X, Y)$ $Corr = \frac{Cov}{\sigma_X \sigma_Y}$
 $Var(X+Y) = Var(X) + Var(Y) + 2Cov(X, Y)$ $-1 \leq Corr \leq 1$
 $Cov(X+Y, W+Z) = Cov(X, W) + Cov(Y, W) + Cov(X, Z) + Cov(Y, Z)$ $-\sigma_X \sigma_Y \leq Cov \leq \sigma_X \sigma_Y$
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$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = \lim_{n \rightarrow \infty} (1 + \frac{x}{n})^n$$

$\phi_X = E(e^{tx}) = \int e^{tx} f(x) dx$
 $E(x^r) = \frac{d^r}{dt^r} \phi_X(t=0)$

Σexp

Gamma: $f(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}$ $\Gamma(\alpha) = \int_0^\infty y^{\alpha-1} e^{-y} dy = (\alpha-1)\Gamma(\alpha-1)$
 $\alpha, \beta > 0$ $E(X) = \alpha\beta$ $Var(X) = \alpha\beta^2$ $\phi_X = \frac{1}{(1+\beta t)^\alpha}$

$\Gamma(1/2) = \sqrt{\pi}$

Exp: $f(x) = \frac{1}{\lambda} e^{-x/\lambda}$ $\phi_X = \frac{1}{1-\lambda t}$ if $t < 1/\lambda$
 $\lambda > 0$ $E(X) = \lambda$ $Var(X) = \lambda^2$

Gamma:
 $M_r = \frac{\beta^r \Gamma(\alpha+r)}{\Gamma(\alpha)}$

$M_r = E(x^r)$
 $M_r = E((x-\mu_1)^r)$
 $\mu'_0 = 1$
 $\mu'_1 = 0$
 $\mu'_2 = Var(X)$

Special case of gamma $\alpha = 1/2$ $\beta = 2$

Chi: $f(x) = \frac{1}{2^{r/2} \Gamma(r/2)} x^{r/2-1} e^{-x/2}$ $\sum_{k=1}^n Z_k^2 = \chi^2_{r=n}$ Z iid
 $X, r > 0$ $E(X) = r$ $Var(X) = 2r$ $\phi_X = (1-2t)^{-r/2}$

If $Y = aX + b$
 $M_Y(t) = e^{bt} M_X(at)$
 If X & Y Indep,
 $Z = X + Y$
 $M_Z = M_X M_Y$

Norm: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$ $\phi_X = e^{t\mu + \frac{1}{2}t^2\sigma^2}$

Geo: $f(x) = p(1-p)^{x-1}$ $\phi_X = \frac{pe^t}{1-(1-p)e^t}$ $E(X) = \frac{1}{p}$ $Var(X) = \frac{1-p}{p^2}$

If $X_1, \dots, X_n$ iid normal rvs then,
 a. $\bar{X}$ and s^2 are indep
 b. $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$

Bin: $f(x) = \binom{n}{x} p^x (1-p)^{n-x}$ $\phi_X = (pe^t + 1-p)^n$ $E(X) = np$ $Var(X) = np(1-p)$

Markov: $P(X \geq a) \leq \frac{E(X)}{a}$ $X \geq 0$ Chebyshev: $P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$ $k > 0$

$\bar{X} = \sum X_i / n$
 $s^2 = \frac{\sum (X_i - \bar{X})^2}{n-1}$
 $E(\bar{X}) = \mu_X$ $Var(\bar{X}) = \frac{\sigma_X^2}{n}$

Order: $g_r(y_r) = \frac{n!}{(r-1)!(n-r)!} \left[\int_0^{y_r} f(x) dx \right]^{r-1} f(y_r) \left[\int_{y_r}^\infty f(x) dx \right]^{n-r}$

Median: $n=2m+1, \tilde{x} = Y_{m+1}$

$h(\tilde{x}) = \frac{(2m+1)!}{m!m!} \left[\int_{-\infty}^{\tilde{x}} f(x) dx \right]^m f(\tilde{x}) \left[\int_{\tilde{x}}^\infty f(x) dx \right]^m$

$n=2m, \tilde{x} = \frac{1}{2}(Y_m + Y_{m+1})$

Mean has less variance than median

$P(\mu - c \leq \bar{X} \leq \mu + c) \geq 1 - \frac{\sigma_X^2}{nc^2}$
 $Z = \frac{\bar{X} - \mu}{\sigma_X / \sqrt{n}}$
 $\lim_{n \rightarrow \infty} M_Z = e^{\frac{1}{2}t^2}$

$\int u dv = uv - \int v du$

$$\lim_{x \rightarrow 0} -\ln(x) - \frac{1}{x} \text{ is } -\infty$$

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Method
of moments

r -th sample moment about origin: $M'_r = \frac{\sum x_i^r}{n}$, $M'_1 = \bar{x}$, $M'_2 = \frac{x_1^2 + \dots + x_n^2}{n}$

By SL of LN, $M'_r \rightarrow E[X^r]$, thus set $M'_r = \mu'_r$ for estimators

MLE: $L(\theta) = f(x_1, x_2, \dots, x_n; \theta)$ Take derivative and set to 0. $\frac{d}{d\theta}$ not $\frac{d}{dx}$
 $\ln(L(\theta))$ maybe easier to calculate.

USE HATS

SL: $\lim_{n \rightarrow \infty} \frac{\sum x_i}{n} = \mu$ with prob 1

CLT: $\lim_{n \rightarrow \infty} \text{Prob}\left(\frac{\sum (x_i - \mu)}{\sqrt{n\sigma^2}} \leq a\right) = \Phi(a)$

WLL: $P\left(\left|\frac{\sum x_i}{n} - \mu\right| \geq \epsilon\right) \rightarrow 0$ as $n \rightarrow \infty$